

LI JING

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. in Physics

2014 - 2020

Dissertation: Physical Symmetry Enhanced Neural Networks

Peking University

Beijing, China

B.S. in Physics

2010 - 2014

B.A. in Economics

2012 - 2014

RESEARCH EXPERIENCE

OpenAI

San Francisco, CA

Researcher

2022-present

- Multimodal representation learning
- Generative models

Meta / Facebook AI Research

New York, NY

Postdoctoral Researcher

2020 - 2022

Advisor: Yann LeCun

- General-purpose and Scalable Self-supervised Learning
 - proposed BarlowTwins, a state-of-the-art scalable self-supervised learning framework for visual representation learning. This work remains one of the highest impact AI research in 2021, which has received 550+ citations and 750+ Github stars.
 - theoretically studied the dimensional collapse problem in contrastive learning, featured in Meta AI Blog and NeurIPS-2021 tutorial.
 - proposed general-purpose self-supervised learning frameworks leveraging equivariance criteria, and universal masking strategy.
 - investigated few-shot alignment methods of multimodal representations
- Self-supervised Learning Applications
 - collaborated with Meta Reality Labs to apply self-supervised learning to EMG devices, and unlocked few-shot gesture recognition applications.
 - improved efficiency in production mobile vision and speech models by applying self-supervised learning and neural architecture search.
 - explored self-supervised learning for protein design.

Massachusetts Institute of Technology

Cambridge, MA

Research Assistant

2016 - 2020

Advisor: Marin Soljacic

- Physical Symmetry Enhanced Neural Networks
 - proposed an efficient unitary recurrent neural network framework based on Givens rotation, which achieved state-of-the-art performances on long-term memory time-series benchmarks. This work has had a broad impact on the optics and quantum computing community.

- proposed gated orthogonal recurrent neural networks, which demonstrated state-of-the-art performances on several NLP applications that requires long-term memory.
- AI for Physics
 - first proposed to use neural networks to assist FDTD simulation, exponentially speed up forward simulation and backup design.
 - proposed transfer learning techniques for photonics simulation, allows data expensive scenarios to leverage easy access data.

ENTREPRENEURIAL EXPERIENCE	Lightelligence Inc. <i>Co-founder, Chief Algorithm Architect</i> MIT Spin-out, Optical AI Accelerator Startup, now C-series	Boston, MA 2018 - 2019
PUBLICATIONS	Google Scholar: https://scholar.google.com/citations?user=VhxDLwcAAAAJ	

Peer Reviewed Machine Learning Conference Papers

1. **Li Jing**, Pascal Vincent, Yann LeCun, Yuandong Tian, “Understanding Dimensional Collapse in Contrastive Self-supervised Learning”, **International Conference on Learning Representations (ICLR)** (2022)
2. Rumen Dangovski, **Li Jing**, Charlotte Loh, Seungwook Han, Akash Srivastava, Brian Cheung, Pulkit Agrawal, Marin Soljagic, “Equivariant Contrastive Learning”, **International Conference on Learning Representations (ICLR)** (2022)
3. Jure Zbontar*, **Li Jing***, Ishan Misra, Yann LeCun, Stéphane Deny, “Barlow Twins: Self-Supervised Learning via Redundancy Reduction”, **International Conference on Machine Learning (ICML)** (2021)
4. Rumen Dangovski, Michelle Shen, Dawson Byrd, **Li Jing**, Desislava Tsvetkova, Preslav Nakov, Marin Soljagic, “We Can Explain Your Research in Layman’s Terms: Towards Automating Science Journalism at Scale”, **AAAI Conference on Artificial Intelligence (AAAI)** (2021)
5. **Li Jing**, Jure Zbontar, Yann LeCun, “Implicit Rank-Minimizing Autoencoder”, **Conference on Neural Information Processing Systems (NeurIPS)** (2020)
6. Matthew Khoury, Rumen Dangovski, Longwu Ou, Yichen Shen, Preslav Nakov, **Li Jing**, “Vector-Vector-Matrix Architecture: A Novel Hardware-Aware Framework for Low-Latency Inference in NLP Applications”, **Conference on Empirical Methods in Natural Language Processing (EMNLP)** (2020)
7. **Li Jing***, Yichen Shen*, Tena Dubček, John Peurifoy, Scott Skirlo, Yann LeCun, Max Tegmark, Marin Soljagic, “Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs”, **Proceedings of International Conference on Machine Learning (ICML)** (2017)

Machine Learning Journal Papers

8. Zhedong Wang, Chao Qian, Tong Cai, Longwei Tian, Zhixiang Fan, Jian Liu, Yichen Shen, **Li Jing**, Jianming Jin, Er-Ping Li, Bin Zheng, Hongsheng Chen,

- “Demonstration of Spider-Eyes-Like Intelligent Antennas for Dynamically Perceiving Incoming Waves”,
Advanced Intelligent Systems (2021)
9. Samuel Kim, Peter Lu, Srijon Mukherjee, Michael Gilbert, **Li Jing**, Vladimir Ceperic, Marin Soljagic, “Integration of Neural Network-Based Symbolic Regression in Deep Learning for Scientific Discovery”,
IEEE Transactions on Neural Networks and Learning Systems (TNNLS) (2020)
 10. Rumen Dangovski*, **Li Jing***, Marin Soljagic, “Rotational Unit of Memory: A Novel Representation Unit for RNNs with Scalable Applications”,
Transactions of the Association for Computational Linguistics (TACL) (2019)
 11. **Li Jing***, Caglar Gulcehre*, John Peurifoy, Yichen Shen, Max Tegmark, Marin Soljagic, Yoshua Bengio, “Gated Orthogonal Recurrent Units: On Learning to Forget”,
Neural Computation (2019)

Physics Journal Papers

12. Thomas Christensen, Charlotte Loh, Stjepan Picek, Domagoj Jakobovic, **Li Jing**, Sophie Fisher, Vladimir Ceperic, John Joannopoulos, Marin Soljagic, “Predictive and generative machine learning models for photonic crystals”,
Nanophotonics (2020)
13. Mihika Prabhu, Charles Roques-Carmes, Yichen Shen, Nicholas Harris, **Li Jing**, Jacques Carolan, Ryan Hamerly, Tom Baehr-Jones, Michael Hochberg, Vladimir Ceperic, John Joannopoulos, Dirk Englund, Marin Soljagic, “A Recurrent Ising Machine in a Photonic Integrated Circuit”,
Optica, (2020)
14. Chao Qian, Bin Zheng, Yichen Shen, **Li Jing**, Erping Li, Lian Shen, Hongsheng Chen, “Deep learning enabled self-adaptive invisibility cloak”,
Nature Photonics (2020)
15. Charles Roques-Carmes, Yichen Shen, Cristian Zanolini, Mihika Prabhu, Fadi Atieh, **Li Jing**, Tena Dubcek, Vladimir Ceperic, John Joannopoulos, Dirk Englund, Marin Soljagic, “Heuristic Recurrent Algorithms for Photonic Ising Machines”,
Nature Communications (2020)
16. Yurui Qu*, **Li Jing***, Yichen Shen, Min Qiu, Marin Soljagic, “Migrating Knowledge between Physical Scenarios based on Artificial Neural Networks”,
ACS Photonics (2019)
17. John Peurifoy, Yichen Shen, **Li Jing**, Yi Yang, Fidel Cano-Renteria, Brendan Delacy, John Joannopoulos, Max Tegmark, Marin Soljagic, “Nanophotonic Particle Simulation and Inverse Design Using Artificial Neural Networks”,
Science Advances (2018)
18. Heng Fan, Yi-Nan Wang, **Li Jing**, Jie-Dong Yue, Han-Duo Shi, Yong-Liang Zhang, Liang-Zhu Mu, “Quantum cloning machines and the applications”,
Physics Reports (2014)
19. Yong-Liang Zhang, Huan Wang, **Li Jing**, Liang-Zhu Mu, Heng Fan, “Fitting magnetic field gradient with Heisenberg-scaling accuracy”,
Scientific Reports (2014)

20. Yong-Liang Zhang, Yi-Nan Wang, Xiang-Ru Xiao, **Li Jing**, Liang-Zhu Mu, VE Korepin, Heng Fan, "Quantum network teleportation for quantum information distribution and concentration", **Physical Review A** (2013)
21. **Li Jing**, Yi-Nan Wang, Han-Duo Shi, Liang-Zhu Mu, Heng Fan, "Minimal input sets determining phase-covariant and universal quantum cloning", **Physical Review A** (2012)
22. Zhao-Xi Xiong, Han-Duo Shi, Yi-Nan Wang, **Li Jing**, Jin Lei, Liang-Zhu Mu, Heng Fan, "General quantum key distribution in higher dimension", **Physical Review A** (2012)
23. Yi-Nan Wang, Han-Duo Shi, **Li Jing**, Zhao-Xi Xiong, Jin Lei, Liang-Zhu Mu, Heng Fan, "Non-compression of quantum phase information", **Journal of Physics A: Mathematical and Theoretical** (2012)
24. Yi-Nan Wang, Han-Duo Shi, Zhao-Xi Xiong, **Li Jing**, Xi-Jun Ren, Liang-Zhu Mu, Heng Fan, "Unified universal quantum cloning machine and fidelities", **Physical Review A** (2011)

Pre-prints

25. Shraman Pramanick*, **Li Jing***, Sayan Nag*, Jiachen Zhu, Hardik Shah, Yann LeCun, Rama Chellappa, "VoLTA: Vision-Language Transformer with Weakly-Supervised Local-Feature Alignment", arXiv: 2210.04135
26. **Li Jing***, Jiachen Zhu*, Yann LeCun, "Masked Siamese ConvNets: Towards an Effective Masking Strategy for General-purpose Siamese Networks", arXiv: 2206.07700
27. Anish Acharya, Sujay Sanghavi, **Li Jing**, Bhargav Bhushanam, Dhruv Choudhary, Michael Rabbat, Inderjit Dhillon, "Positive Unlabeled Contrastive Learning", arXiv: 2206.01206
28. Andrew Ma, Yang Zhang, Thomas Christensen, Hoi Chun Po, **Li Jing**, Liang Fu, Marin Soljagic, "Topogivity: A Machine-Learned Chemical Rule for Discovering Topological Materials ", arXiv: 2202.05255
29. Ileana Rugina, Rumen Dangovski, Renbin Liu, **Li Jing**, Preslav Nakov, Marin Soljagic, "Data-Informed Global Sparseness in Attention Mechanisms for Deep Neural Networks", arXiv: 2012.02030
30. Evan Vogelbaum, Rumen Dangovski, **Li Jing**, Marin Soljačić, "Contextualizing Enhances Gradient Based Meta Learning", arXiv: 2007.10143
31. **Li Jing***, Rumen Dangovski*, Marin Soljagic, "WaveletNet: Logarithmic Scale Efficient Convolutional Neural Networks for Edge Devices", arXiv:1811.11644
32. Han-Duo Shi, Yi-Nan Wang, **Li Jing**, Ru-Quan Wang, Liang-Zhu Mu, Heng Fan, "Quantum key distribution based on a quantum retrodiction protocol", arXiv: 1205.3556

Patents

33. Yichen Shen, **Li Jing**, Rumen Dangovski, Peng Xie, Huaiyu Meng, Matthew Khoury, Cheng-Kuan Lu, Ronald Gagnon, Maurice Steinman, Jianhua Wu, Arash Hosseinzadeh, "Optoelectronic computing systems"
US patent number 11,281,972, issued in March 2022
34. Charles Roques-arnes, Yichen Shen, **Li Jing**, Tena Dubcek, Scott A Skirlo, Hengameh Bagherianlemraski, Marin Soljagic, "Optical Ising machines and optical convolutional neural networks"
US patent number 11,017,309, issued in May 2021

INVITED TALKS

1. Towards Scalable and General-purpose Self-supervised Learning
Adept AI San Francisco, CA, 2022
2. Towards Scalable and General-purpose Self-supervised Learning
Cohere AI Virtual, 2022
3. Towards Scalable Self-supervised Learning
Salesforce Research Virtual, 2022
4. Understanding and Solving Dimensional Collapse in Contrastive SSL
Flatiron Research Institute New York, NY, 2022
Center for Computational Mathematics
5. Understanding and Solving Dimensional Collapse in Contrastive SSL
UC Davis Virtual, 2022
AI, Robotics and Automation Seminar
6. Implicit Regularization and Self-supervised Learning
OpenAI Virtual, 2021
7. On Dimensional Collapse in Contrastive Self-supervised Learning
Google Research Virtual, 2021
8. Barlow Twins: Self-supervised Learning via Redundancy Reduction
International Conference on Machine Learning (ICML) Virtual, 2021
9. Implicit Rank-Minimizing Autoencoder
Conference on Neural Information Processing Systems (NeurIPS) Virtual, 2020
10. Physics Inspired Deep Learning Algorithms
Facebook AI Research New York, NY, 2019
11. Phase-encoded Recurrent Neural Networks and their Applications
DeepMind Mountain View, CA, 2019
12. Phase-encoded RNNs and Scientific Text Summarization
Amazon Alexa AI Cambridge, MA, 2019
13. Photonics and AI: Algorithms and Applications
APS Physics of AI conference Yorktown Heights, NY, 2019
14. **MIT-CHIEF Annual Conference, AI panel** Cambridge, MA, 2018
15. Nanophotonic Particle Simulation and Inverse Design Using Artificial Neural Networks
Conference on Neural Information Processing Systems (NIPS)
Deep Learning for Physical Science Workshop Long Beach, CA, 2018

16. Gated Orthogonal Recurrent Units: On Learning to Forget
AAAI Conference on Artificial Intelligence (AAAI)
 Workshop on Reasoning and Learning for Human-Machine Dialogues
 New Orleans, LA, 2018
17. Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs
International Conference on Machine Learning (ICML)
 Sydney, Australia, 2017

PRESS COVERAGE

1. Understanding and mitigating dimensional collapse in contrastive learning methods
Meta AI Blog 2022
2. Self-Adaptive Invisibility Cloak: The “Magic” in Optics.
Techacute 2020
3. Solving complex problems at the speed of light.
SciTechDaily 2020
4. Lightelligence releases prototype of its optical AI accelerator chip
Venturebeat 2019
5. Can science writing be automated? A neural network can read scientific papers and render a plain-English summary.
MIT News 2019
6. AI-based method could speed development of specialized nanoparticles. Neural network could expedite complex physics simulations.
MIT News 2018

ACADEMIC SERVICES

Conference reviewer

- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- Conference on Neural Information Processing Systems (NeurIPS)
- Conference on Empirical Methods in Natural Language Processing (EMNLP)

Journal reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Computational Intelligence Magazine (CIM)
- IEEE Photonic Technology Letters
- Proceedings of the National Academy of Sciences (PNAS)
- Physical Review Letters
- Physical Review A
- Physical Review B
- Physical Review X
- Physical Review Research
- Physical Review Applied
- Scientific Reports
- Journal of Applied Physics

- Optics Express
- Optica
- Advanced Photonics
- Nanophotonics
- ACS Photonics
- Nature Photonics

MENTORING

- | | |
|---|------------|
| 1. Shraman Pramanick
Ph.D at JHU, intern at Meta | 2022 |
| 2. Ruogu Lin
Ph.D at CMU, intern at Meta | 2022 |
| 3. Anish Acharya
Ph.D at UT Austin, intern at Meta | 2021, 2022 |
| 4. Jiachen Zhu
Ph.D at NYU, intern at Meta / Facebook AI Research | 2021, 2022 |
| 5. Pawan Goyal
Undergrad at MIT | 2019 |
| 6. Lay Jain
Undergrad at MIT, now Quantitative Researcher at Two Sigma | 2019 |
| 7. Alvaro Inesta
Exchange Master student at MIT, now PhD at TU Delft | 2017, 2018 |
| 8. Ivan Ivanov
RSI program at MIT, now undergrad at Cambridge University | 2017 |
| 9. Rawn Henry
Undergrad at MIT, now AI Engineer at Nvidia | 2017 |
| 10. John Peurifoy
Undergrad at MIT, now CEO at Floating point group | 2016, 2017 |
| 11. Rumen Dangovski
Ph.D student at MIT EECS, intern at Meta | 2017-2022 |

SELECTED HONORS AND AWARDS

- | | |
|--|------|
| • Forbes China 30 under 30 - Enterprise Technology | 2019 |
| • MIT Presidential Fellowship | 2014 |
| • Peking University Outstanding Graduate | 2014 |
| • Peking University Academic Innovation Award | 2013 |
| • International Physics Olympiad (IPhO) - Gold Medal | 2010 |